

Klaus Havelund, PhD

Senior Research Scientist and Principal

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Profile

Senior Research Scientist and Principal at NASA JPL with 30+ years in formal methods, runtime verification, and automated software analysis. 189 publications, 14,155 citations, h-index: 54.

Education

Ph.D. in Computer Science — University of Copenhagen, Denmark 1990–1994
PhD carried out in part at École Normale Supérieure, Paris
Master in Computer Science — University of Copenhagen, Denmark 1986

Professional Experience

NASA Jet Propulsion Laboratory (JPL), California 2006–Present
Senior Research Scientist (2009–Present) — Principal (2007–Present)
Kestrel Technology, California 2005–2006
Researcher
NASA Ames Research Center, California 1997–2005
Researcher

Earlier Research Positions

Aalborg University (1996–97), **Ecole Polytechnique and Paris 6 University** (1994–96), **DDC¹/CRI, Denmark** (1984–90)

Student Programming Jobs

Skibsteknisk Laboratorium, RC Datacenter, Christian Rovsing (1977–1981)

Projects

Current JPL Projects

PyContract: Python library for runtime verification, used on Endurance rover for execution analysis and Europa Clipper session report analysis — **Daut:** Scala log analysis library (precursor to PyContract), used on Endurance — **Fuzz:** Python constraint-based tool for generating randomized command sequences with temporal logic constraints, applied to Europa Clipper FIT testing — **TaskSAT:** Python tool using constraint solving for verifying tasknets (MEXEC inputs), developed for Endurance project as part of PAAS strategic RTD — **JPL Coding Standards:** C++ (with Robert Bochino) and Python standards — **LLM Evaluation:** Evaluation of LLM-based programming tools (Windsurf, Claude Code), including F Prime application development — **Python Model Checker:** Developed with GenAI and University of Manchester as part of PAAS strategic RTD — **Other Research:** Combinatorial testing for Caleb Wagner’s testing framework, Autonomica testing framework, STest implementation in Python, LLM-based log analysis

Past JPL Projects

Java Coding Standard (2014) — **DejaVu:** First-order past time temporal logic monitoring with BDDs — **nfer:** Logic and system for inferring interval abstractions — **K Language:** Constraint-based specification language for SysML class diagrams, developed with Bradley Clement — **TraceContract:** Scala log analysis API, used by NASA Ames during Lunar LADEE mission — **LogFire:** Scala implementation of RETE algorithm for runtime verification, used by Rajeev Joshi for Curiosity rover telemetry — **LogScope:** Python tool for checking log files against temporal logic specifications — **LogScope/C++:** Monitoring DSL with translation to C++ — **Armor:** CIL-based tool for instrumenting and monitoring C programs — **ScalaFM:** Exploring Scala for modeling (hierarchical state charts) — **Mcai 2.0:** Model Checking and Abstract Interpretation, NSF Expeditions project — **Ruler:** Runtime verification logic — **Survivable Software:** GCC-based framework for static analysis, instrumentation, and monitoring — **Lscs:** Launch Site Command and Control System for space shuttle — **Map:** Translator from ASPEN planning language to Promela — **FPP State Machine DSL:** Contributions to F Prime sub-language for state machines

Past NASA Ames Projects

Java PathFinder (JPF): Pioneered and developed the first version of this model checker for Java programs

¹https://en.wikipedia.org/wiki/Dansk_Datamatik_Center

Awards & Recognition

2021: Festschrift Workshop (Formal Methods in Outer Space) at ISO LA 2021 for 65th birthday — **2020:** SIGSOFT Impact Paper Award, JPL Magellan Award (\$10K) — **2018:** RV Test of Time Award, Best Benchmark Award (DejaVu benchmark) — **2017:** JPL Voyager Award — **2016:** ASE Most Influential Paper — **2015:** CRV Competition Win (LogFire) — **2014:** ASE Most Influential Paper — **2011:** RV Best Paper, JPL Mariner Award — **2010:** JPL Ranger Award — **2009:** JPL Mariner Award, FLC Outstanding Technology Award — **2008–09:** Royal Academy Fellowship (UK) — **2008:** ACM Distinguished Paper, NASA Group Achievement — **2006:** NASA Tech Brief Award — **2003:** NASA TGIR Award — **2002:** EASST Best Paper

Invited Presentations

ASM'03 (Italy, 2003) — Joint CAV/ISSTA Special Event on Specification, Verification, and Testing of Concurrent Software (Boston, 2004) — Danish industry via CISS (Aalborg, 2004) — CASSIS'05 (Nice, 2005) — VVEIS'05 (Miami, 2005) — 8th JPL-GSFC Quality Mission Software Workshop (Santa Barbara, 2006) — PADTAD'06 (Portland, 2006) — ARTIST2 Summer School (Autrans, France, 2008) — Talk for 150 10-11 graders (Manchester, UK, 2009) — COMPASS'09 (York, UK, 2009) — 9th RV Workshop (Grenoble, 2009) — Software Reliability for Space Missions (Pasadena, 2009) — FMA'09 (Eindhoven, 2009) — EMSOFT'11 (Taipei, 2011) — SEFM'11 (Montevideo, 2011) — TORRENTS'11 (Toulouse, 2011) — HOWARD-60 (Manchester, 2011) — NFM'12 Panel (Norfolk, 2012) — SSS'12 (Minneapolis, 2012) — Marktoberdorf Summer School (2012) — LCCC workshop (Lund, 2012) — ISO LA'12 (Crete, 2012) — ICTSS'12 (Aalborg, 2012) — WODA'13 (Houston, 2013) — CERIST Autumn School (Algiers, 2013) — IFIP WG 1.9/2.15 Observer (Orlando, 2013; Vienna, 2014) — ISO LA'14 (Corfu, 2014) — TASE'14 (Changsha, 2014) — ISO LA'14 Tool Panel (Corfu, 2014) — FTSCS'14 (Luxembourg, 2014) — EITEC'15 (Seattle, 2015) — SyDe Summer School (Bremen, 2015) — IFIP WG 2.3 Observer (Pasadena, 2016) — CIT 2017 (Detroit, 2016) — RV 2016 (Madrid, 2016) — USC CS Colloquium (Los Angeles, 2016) — NFM'16 Panel (Minneapolis, 2016) — Software Correctness and Reliability Workshop (ETH Zurich, 2017) — SPIN 2018 with Doron Peled (Malaga, 2018) — ISO LA 2018 (Limassol, 2018) — ISO LA 2018 RV-TheToP Track with Doron Peled (Limassol, 2018) — RV 2018 with Grigore Rosu (Limassol, 2018) — ATVA 2020 with Doron Peled (Hanoi, virtual, 2020) — ESEC/FSE 2020 with Willem Visser (Sacramento, virtual, 2020) — ICTAC 2020 (Macau, virtual, 2020) — ISO LA 2022 (Rhodes, 2022) — JanFest 2023 invited paper (Bremen, 2023) — VORTEX 2024 (Vienna, 2024, presented by Doron Peled) — ICTAC 2025 (Marrakesh, 2025) — OSDF 2026 (JPL, 2026) — RE×AI 2026 (Los Angeles, 2026)

Academic Services

Conference Organization

Co-founder of Runtime Verification (RV) conference series (started 2001, became conference 2010) — RV 2001, 2002, 2004, 2006, 2010, 2011, 2014, 2016, 2017, 2021, RVCASE'25 — SPIN 2000, 2008, 2017 — NFM 2011, 2015, 2022, 2026 (18th NFM at USC) — TACAS 2014 — VVPS workshops 2005, 2009, 2011 — Dagstuhl seminars on Runtime Verification 2007, 2010, 2014, 2017 — CPS School 2014 — ISO LA/AISO LA tracks: ISO LA 2016 (2 tracks), ISO LA 2018, ISO LA 2021, ISO LA 2022, AISO LA 2023, AISO LA 2024, AISO LA 2025, AISO LA 2026 — FM 2018 Industry Day

Program Committees

258 program committee memberships across 105 unique conferences/workshops: ABZ, AMDE, AOSD, ASARTI, ATVA, CAV, CONFENIS, DAW, DRHE, ECOOP, EITEC, Europar, FASE, FATES, FM, FMFun, FMI, FMICS, FMOODS & FORTE, FMPPTA, FMSPLE, FORMABS, FORTE, FTSCS, HAS, HSCC, HVC, I-Day@FM, ICAPS, ICFEM, ICI, ICSE, ICSEA, ICTAC, ICTSS, IEEE SMC-IT/SCC, INTELLI, ISSRE, ISO LA, InterAVT, Jan65, JPF, K, MOCHAP, MODELSWARD, MSVVEIS, MUSEPAT, MVI, MoChArt, Modularity, NFM, PADTAD, PATV, PLDI, PPMG, PrePost, REOS, RTT, RV, SAC-SVT, SATE, SAW, SEFM, SEW, SFEDL, SHM, SLE, SMC-IT, SPIN, SSV, SafePlan, Scala Days, ScottFest, SummerSim, SYNASC, TACAS, TAP, TASE, VMCAI, VORTEX, VSTTE, VVEIS, VVPS, WODA, iFM

Steering Committees

Runtime Verification (RV) conference (Chair 2010–2017, Member 2017–present) — NASA Formal Methods (NFM) symposium (Chair 2011–2017, Member 2017–present) — FME Advisory Committee — ETAPS (2014–2015) — SPIN (2009)

Editorial

FoMaC: Transactions on Foundations for Mastering Change

Other Service

Chair of the Formal Methods Europe (FME) Industry Committee — JPL's SRS (Senior Research Scientist) Council (served twice) — Member of undisclosed JPL review committees — Lecturer at annual F Prime workshops (2022–2023)

Memberships

ACM (Association for Computing Machinery), **IFIP WG 1.9/2.15** (Verified Software)

Publications

Complete List (189 publications — 14,155 citations — h-index: 54)

Books and Book Chapters (4)

1. The RAISE Language Group. *The RAISE Specification Language*. Prentice Hall, 1992. (Havelund wrote tutorial section, 250 pages)
2. Klaus Havelund. *RAISE in Perspective*. Chapter in *Logics of Specification Languages*, Springer, 2007.
3. Klaus Havelund, Giles Reger, Daniel Thoma, Eugen Zălinescu. *Monitoring Events that Carry Data*. Chapter in *Lectures on Runtime Verification*, LNCS 10457, Springer, 2018.
4. Ylies Falcone, Klaus Havelund, Giles Reger. *A Tutorial on Runtime Verification*. NATO Science for Peace and Security Series, IOS Press, 2013.

Journal Papers (38)

1. E. Ábrahám, K. Havelund. Some Recent Advances in Automated Analysis. *STTT*, 18(2):121-128, 2016.
2. R. Agarwal et al. Detection of Deadlock Potentials in Multithreaded Programs. *IBM J. Research & Development*, 54(5), 2010.
3. C. Artho et al. Combining Test-Case Generation and Runtime Verification. *J. Theoretical Computer Science*, 336(2-3), 2005.
4. C. Artho, K. Havelund, A. Biere. High-Level Data Races. *Software Testing, Verification and Reliability*, 13(4), 2004.
5. H. Barringer, A. Groce, K. Havelund, M. Smith. Formal Analysis of Log Files. *J. Aerospace Computing, Info., and Comm.*, 7(11):365-390, 2010.
6. H. Barringer, D. Rydeheard, K. Havelund. Rule Systems for Run-Time Monitoring. *J. Logic and Computation*, 20(3), 2008.
7. E. Bartocci, Y. Falcone et al. First Intl. Competition on Runtime Verification. *STTT*, 2017.
8. M. Bennett, R. Borgen, K. Havelund, M. Ingham, D. Wagner. Prototyping a DSL for Monitor and Control Systems. *J. Aerospace Computing*, 7(11):338-364, 2010.
9. S. Bensalem, K. Havelund, A. Orlandini. Verification & Validation Meets Planning & Scheduling. *STTT*, 16(1), 2014.
10. D. Bjørner, A. Haxthausen, K. Havelund. Formal Model-oriented Software Development Methods. *Future Generation Computer Systems*, 7, 1992.
11. D. Bjørner, K. Havelund. A Half Century of Formal Methods. Submitted, under review, 2021.
12. E. Bodden, K. Havelund. Aspect-oriented Race Detection in Java. *IEEE Trans. Software Engineering*, 36(4):509-527, 2010.
13. G. Brat et al. Experimental Evaluation of V&V Tools on Martian Rover Software. *Formal Methods in System Design*, 25(2), 2004.
14. M. Broy et al. Does Every Computer Scientist Need to Know Formal Methods? *Formal Aspects of Computing*, 2024 (to appear).
15. J. Deshmukh, K. Havelund, I. Perez. Introduction to Selected Papers from NFM 2022. *Innovations in Systems and Software Engineering*, 2024.
16. H. Erdogmus, K. Havelund. Introduction to Selected Papers from SPIN 2017. *STTT*, 21, 2019.
17. Y. Eytani, K. Havelund, S. Stoller, S. Ur. Framework and Benchmark for Testing Tools for Multi-Threaded Programs. *J. Concurrency and Computation*, 2005.
18. C. George, K. Havelund, M. Nielsen, K.R. Wagner. The RAISE Language, Method and Tools. *Formal Aspects of Computing*, 1(1), 1989.
19. A. Groce, K. Havelund, G. Holzmann, R. Joshi, R-G. Xu. Establishing Flight Software Reliability. *Annals of Math. and AI*, accepted.
20. K. Havelund. Rule-based Runtime Verification Revisited. *STTT*, 17(2):143-170, 2015.
21. K. Havelund, G. Holzmann. Programming Event Monitors. *STTT*, 26:33-47, 2023.
22. K. Havelund, R. Kumar. Verified Change. *FoMaC Transactions*, submitted 2016.
23. K. Havelund, K.G. Larsen. The Fork Calculus. *Nordic J. Computing*, 1, 1994.
24. K. Havelund, M. Lowry, J. Penix. Formal Analysis of a Space Craft Controller using SPIN. *IEEE Trans. Software Eng.*, 27(8):749-765, 2001.
25. K. Havelund, D. Peled. An Extension of LTL with Rules. *STTT*, 23, 2021.
26. K. Havelund, D. Peled, D. Ulus. First-Order Temporal Logic Monitoring with BDDs. *FMSD*, 56:1-21, 2020.
27. K. Havelund, D. Peled. On Monitoring Linear Temporal Properties. *FMSD*, 60:405-425, 2023.
28. K. Havelund, T. Pressburger. Model Checking Java Programs using Java PathFinder. *STTT*, 2(4):366-381, 2000.
29. K. Havelund, G. Roşu. Overview of Runtime Verification Tool Java PathExplorer. *FMSD*, 24(2), 2004.
30. K. Havelund, G. Roşu. Efficient Monitoring of Safety Properties. *STTT*, 6(2):158-173, 2004.
31. K. Havelund, W. Visser. Program Model Checking as a New Trend. *STTT*, 4(1), 2002.
32. S. Kauffman, K. Havelund, R. Joshi, S. Fischmeister. Inferring Event Stream Abstractions. *FMSD*, 2018.
33. S. Kauffman, K. Havelund, S. Fischmeister. What Can We Monitor Over Unreliable Channels? *STTT*, 23, 2021.
34. G. Roşu, K. Havelund. Rewriting-Based Techniques for Runtime Verification. *Intl. J. Automated Software Eng.*, 12(2):151-197, 2005.
35. J. Seyster et al. InterAspect: Aspect-Oriented Instrumentation with GCC. *FMSD*, 41(3):295-320, 2012.
36. N. Shafiei, K. Havelund, P. Mehrlitz. Concurrent Runtime Verification of Data Rich Events. *STTT*, 2023.
37. O. Sokolsky, K. Havelund, I. Lee. Introduction to Special Section on Runtime Verification. *STTT*, 14(3):243-247, 2012.
38. W. Visser, K. Havelund, G. Brat, S. Park, F. Lerda. Model Checking Programs. *Intl. J. Automated Software Eng.*, 10(2), 2003.

Conference and Workshop Papers (147)

1. W. Ahrendt, B.K. Aichernig, K. Havelund. AI Assisted Programming (Track Intro). *AISoLA 2024*, LNCS.
2. W. Ahrendt, K. Havelund. AI Assisted Programming (Track Intro). *ISoLA 2023*, LNCS 14380.
3. W. Ahrendt, B. Aichernig, K. Havelund. AI Assisted Programming (Track Intro). *AISoLA 2025*, LNCS (to appear).
4. W. Ahrendt, B. K. Aichernig, K. Havelund. AI Assisted Programming (AISoLA 2025 Post-Proceedings Track Introduction). *AISoLA 2025*, post-proceedings.
5. W. Ahrendt, B. K. Aichernig, K. Havelund. AI Assisted Programming (AISoLA 2026 Track Introduction). *AISoLA 2026*.
6. B. Aichernig, K. Havelund. AI-Assisted Programming with Test-based Refinement. *ISoLA 2023*, LNCS 14380.
7. B. Aichernig, K. Havelund. Correct-ish by Design. *ISoLA/AISoLA 2024*, LNCS.
8. J. Anton et al. Towards Industrial Scale Development of Custom Static Analyzers. *Static Analysis Summit*, NIST, 2006.
9. C. Artho et al. Experiments with Test Case Generation and Runtime Analysis. *ASM'03*, LNCS 2589, 2003.
10. C. Artho, K. Havelund, S. Honiden. Visualization of Concurrent Program Executions. *SACT'07*, 2007.
11. C. Artho, K. Havelund. Applying Jlint to Space Exploration Software. *VMCAI'04*, LNCS 2937, 2004.
12. C. Artho, K. Havelund, A. Biere. High-Level Data Races. *VVEIS'03*, 2003.

13. C. Artho, K. Havelund, A. Biere. Using Block-Local Atomicity to Detect Stale-Value Errors. ATVA'04, 2004.
14. C. Artho, K. Havelund, R. Kumar, Y. Yamagata. Domain-Specific Languages with Scala. ICFEM 2015, LNCS 9407, 2015.
15. H. Barringer, Y. Falcone, K. Havelund, G. Reger, D. Rydeheard. Quantified Event Automata. FM 2012, LNCS 7436.
16. H. Barringer, A. Goldberg, K. Havelund, K. Sen. Program Monitoring with LTL in Eagle. PADTAD'04, 2004.
17. H. Barringer, A. Goldberg, K. Havelund, K. Sen. Rule-Based Runtime Verification. VMCAI'04, LNCS 2937, 2004.
18. H. Barringer, A. Groce, K. Havelund, M. Smith. Entry Point for Formal Methods. FMA'09, 2009.
19. H. Barringer, K. Havelund. TraceContract: A Scala DSL for Trace Analysis. FM'11, LNCS 6664, 2011.
20. H. Barringer, K. Havelund. Internal vs External DSLs for Trace Analysis. RV 2011, 2011.
21. H. Barringer, K. Havelund, E. Kurklu, R. Morris. Checking Flight Rules with TraceContract. Scala Days 2011.
22. H. Barringer, K. Havelund, D. Rydeheard, A. Groce. Rule Systems for Runtime Verification: Tutorial. RV'09, LNCS 5779, 2009.
23. H. Barringer, D. Rydeheard, K. Havelund. Rule Systems for Run-Time Monitoring. RV'07, LNCS 4839, 2007.
24. M. Bennett, R. Borgen, K. Havelund, M. Ingham, D. Wagner. Development of Prototype DSL for M&C Systems. IEEE Aerospace, 2008.
25. S. Bensalem, J-C. Fernandez, K. Havelund, L. Mounier. Confirmation of Deadlock Potentials. PADTAD'06, 2006.
26. S. Bensalem, K. Havelund. Dynamic Deadlock Analysis of Multi-Threaded Programs. PADTAD'05, LNCS 3875, 2005.
27. D. Bjørner, K. Havelund. 40 Years of Formal Methods. FM 2014, LNCS 8442, 2014.
28. E. Bodden, K. Havelund. Racer: Effective Race Detection Using AspectJ. ISSTA'08, 2008. **[ACM Distinguished Paper]**
29. G. Brat et al. Experimental Evaluation of V&V Tools on Martian Rover Software. CMU/SEI Workshop, 2003.
30. M. Broy, K. Havelund, R. Kumar. Towards a Unified View of Modeling and Programming. ISoLA 2016, LNCS 9952, 2016.
31. M. Broy, K. Havelund, R. Kumar, B. Steffen. Unified View (Track Summary). ISoLA 2016, LNCS 9952, 2016.
32. M. Broy, K. Havelund, R. Kumar, B. Steffen. Unified View (Track Intro). ISoLA 2018, LNCS (TBD).
33. W. Buntine et al. Transformation Systems at NASA Ames. Software Transformation Systems, 1999.
34. I. Cohen, K. Havelund, M. Omer, D. Peled. Monitoring LLM Conversations with First-Order Temporal Logic. RV 2026.
35. I. Cohen, K. Havelund, M. Omer, D. Peled. DejaVuGuard: Runtime Verification of LLM Conversations with First-Order Temporal Logic. RV 2026. Tool paper.
36. I. Cohen, K. Havelund, M. Omer, D. Peled. Temporal Guardrails for LLM Conversations: A Runtime Verification Framework. SAIV 2026, LNCS, pp. 145–166.
37. I. Cohen, K. Havelund, D. Peled, Y. Goldberg. Power of Reframing: Using LLMs in Synthesizing RV Monitors. RV 2025, LNCS 16087, pp. 202–212.
38. I. Cohen, K. Havelund, D. Peled, Y. Goldberg. From Natural Language to Monitors via LLM-Assisted Disambiguation. RV 2025.
39. M. d'Amorim, K. Havelund. Event-Based Runtime Verification of Java Programs. WODA'05, 2005.
40. D. Dams, K. Havelund, S. Kauffman. A Python Library for Trace Analysis. RV 2022, LNCS 13498, 2022.
41. D. Dams, K. Havelund, S. Kauffman. Runtime Verification as Documentation. ISoLA 2022, LNCS 13702, 2022.
42. D. Drusinsky, K. Havelund. Execution-Based Model Checking of Interrupt-Based Systems. DSN'03 Workshop, 2003.
43. B. Duckett, K. Havelund, L. Stewart. Space Telemetry Analysis with PyContract. JanFest 2023, LNCS 14165, 2023.
44. M.S. Feather, S.L. Cornford, K. Havelund. Assurance of Model-Based Autonomy. RAMS 2022, 2022.
45. R.E. Filman, K. Havelund. Realizing Aspects by Transforming for Events. ASE'02, 2002.
46. R.E. Filman, K. Havelund. Source-Code Instrumentation and Quantification. Workshop on Foundations of AOP, 2002.
47. A. Goldberg, K. Havelund. Automated Runtime Verification with MEXICO. RV'05, ENTCS 144, 2005.
48. A. Goldberg, K. Havelund. Instrumentation of Java Bytecode for Runtime Analysis. FATES'05, 2005.
49. K. Havelund. Using Runtime Analysis to Guide Model Checking of Java Programs. SPIN'00, LNCS 1885, 2000.
50. K. Havelund. Monitoring with Data Automata. ISoLA 2014, LNCS 8803, 2014.
51. K. Havelund. Rule-Based Runtime Verification of Cyber-Physical Systems. ISoLA 2014, LNCS 8803, 2014.
52. K. Havelund. Closing the Gap between Specification and Programming. ISoLA 2016, LNCS 9952, 2016.
53. K. Havelund. Data Automata in Scala. ICTAC 2014, LNCS 8687, 2014.
54. K. Havelund. Rethinking Model Checking of Java Programs. FM 2018, LNCS 10951, 2018.
55. K. Havelund. Monitoring the Execution of Space Craft Flight Software. ISoLA 2018, LNCS 11245, 2018.
56. K. Havelund. The Many Roles of Runtime Verification. RV 2021, LNCS 12974, 2021.
57. K. Havelund, P. Alexander. If-Then-Else Theorems for Sets. Festschrift Klaus Grue, LNCS 7850, 2013.
58. K. Havelund, C. Artho, V. Jagannathan et al. What's Not in Your Benchmark? DaCapo Concurrency. PADTAD'10, 2010.
59. K. Havelund, S. Bensalem. Models of Behavior Composition (Track Intro). ISoLA 2010, LNCS 6415, 2010.
60. K. Havelund, I. Becerra, M.S. Feather. Three Mission Scenarios for Data Analysis Using Model Checking. NFM 2024, LNCS 14627, 2024.
61. K. Havelund, M. Broy, R. Kumar, B. Steffen. Revisiting Unified View 10 Years Later. ISoLA 2025, LNCS (to appear).
62. K. Havelund, M.S. Feather, I. Perez. Temporal Logic Anomaly Detection. VORTEX 2023, 2023.
63. K. Havelund, A. Goldberg. Verify Your Runs. VMCAI'05, LNCS 3385, 2005.
64. K. Havelund, G. Holzmann. Effectiveness of Randomization in Spin. SPIN'09, LNCS 5578, 2009.
65. K. Havelund, R. Joshi. Checking Flight Rules During Anomaly Resolution. FMA'11, 2011.
66. K. Havelund, R. Joshi. Using Aspect-Oriented Programming to Extend JPF. SPIN'08, LNCS 5156, 2008.
67. K. Havelund, R. Joshi, C. Artho. First-Order Runtime Verification of Multithreaded Java Programs. SPIN'09, LNCS 5578, 2009.
68. K. Havelund, B. Johnson. Algorithms for Garbage Collection With Multiple Generations. VDM-Europe'91, LNCS 551, 1991.
69. K. Havelund, R. Kumar. Outline of a Specification Language. ISoLA 2016, LNCS 9953, 2016.
70. K. Havelund, R. Kumar. The K Language. WOET'14, 2014.
71. K. Havelund, R. Kumar, D. Smith. Flight Rule Verification by Model Checking. MOCHAP 2015, EPTCS 187, 2015.
72. K. Havelund, K.G. Larsen. The Fork Calculus. ICALP'93, LNCS 700, 1993.
73. K. Havelund, G. Liu, C. Pasareanu. Automated Event Stream Analysis with Prefix Tree Acceptors. ISoLA 2018, LNCS 11247, 2018.
74. K. Havelund, G. Liu, C. Pasareanu. Model Inference of Event Streams. ISoLA 2020, LNCS 12476, 2020.
75. K. Havelund, D. Peled. Efficient Runtime Verification of First-Order Temporal Properties. SPIN'18, LNCS 10869, 2018.
76. K. Havelund, D. Peled. BDD-Based Runtime Verification. ISoLA 2018, LNCS 11247, 2018.
77. K. Havelund, D. Peled. Measuring First-Order Temporal Logic Monitors. ISoLA 2020, LNCS 12478, 2020.
78. K. Havelund, D. Peled. An Extension of LTL with Rules and its Application to RV. FMCAD 2020, 2020.
79. K. Havelund, D. Peled, D. Ulus. First-Order Temporal Logic Monitoring with BDDs. FMCAD 2017, 2017.
80. K. Havelund, D. Peled, D. Ulus. DejaVu Benchmark. RV 2018, LNCS 11237, 2018. **[Best Benchmark Award]**
81. K. Havelund, A. Pnueli. Programs, Protocols and Timing Diagrams. ECCC'97, LNCS 1414, 1997.
82. K. Havelund, T. Pressburger. Model Checking Java Programs using Java PathFinder. STTT Special Issue, 2000.
83. K. Havelund, G. Roşu. Monitoring Java Programs with Java PathExplorer. RV'01, ENTCS 55(2), 2001. **[RV 2018 Test of Time]**
84. K. Havelund, G. Roşu. Synthesizing Monitors for Safety Properties. TACAS'02, LNCS 2280, 2002. **[EASST Best Paper]**
85. K. Havelund, G. Roşu. Monitoring Programs using Rewriting. ASE'01, 2001. **[ASE 2016 Most Influential]**
86. K. Havelund, G. Roşu. Testing Linear Temporal Logic Formulae on Finite Execution Traces. PDMC'01, 2001.
87. K. Havelund, G. Roşu. Runtime Verification Workshop Series. RV'04, 2004.
88. K. Havelund, N. Shankar. Experiments in Theorem Proving and Model Checking for Protocol Verification. FME'96, LNCS 1051, 1996.
89. K. Havelund, M. Sirjani, F. Arbab, E.B. Johnsen. Monitoring Actors and Objects. ISoLA 2012, LNCS 7610, 2012.
90. K. Havelund, L. Stewart, N. Shafiei. Analyzing Space Flight Logs with TraceContract. SMC-IT/SCC 2023, 2023.
91. K. Havelund, W. Visser. Program Model Checking as a New Trend. STTT, 2002.
92. K. Havelund, W. Visser. Dynamic Partial Order Reduction for Model Checking Software. SPIN'00, LNCS 1885, 2000.
93. K. Havelund et al. Formal Analysis of the Remote Agent Before and After Flight. SPIN'00, LNCS 1885, 2000.

94. G. Holzmann, K. Havelund. Checking Code Generated by MDD Tools. ISO/LA 2012, LNCS 7609, 2012.
95. M. Ingham, K. Havelund. Challenges in Verifying Autonomous Systems. AIAA Space 2007, 2007.
96. M. Ingham, K. Havelund, A. Barrett, R. Rasmussen. Issues in Runtime Assertion Checking. MOCHAP 2012, AAAI Technical Report, 2012.
97. R. Joshi, K. Havelund. Towards a Framework for Analyzing Fault Diagnostics. FMA'12, 2012.
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Teaching

Most of my career has been spent in research labs, so my teaching experience is limited. Appointed Lecturer in Computer Science at California Institute of Technology: taught classes on program monitoring and runtime verification (2008, 2009).